

Soluciones Examen Final

M. Siddikov

July 7, 2025

Problem 1 - Homework 1

- ① (20 points) A photon (γ) of energy E_γ scatters inelastically on the proton which rests in the laboratory frame and produces a heavy neutral J/ψ meson via $\gamma p \rightarrow J/\psi p$ process. The masses of the J/ψ meson and proton are given by M and m respectively. The recoil proton emerges at angle α_1 with respect to collision axis. The J/ψ meson almost immediately decays into a pair of oppositely charged muons of equal masses, $J/\psi \rightarrow \mu^+ \mu^-$.

- ① (4 points) Find the energies of the produced J/ψ meson and recoil proton and their relative velocity v_{rel} .
- ② (4 points) Assuming that the J/ψ meson decays isotropically in its rest frame, find the minimal and maximal energies of the muons in the laboratory frame.
- ③ (4 points) Find the minimal photon energy E_γ which is required to initiate this process.
- ④ (4 points) Write out explicit expressions for the Mandelstam variables s, t, u in terms of energies and the angle α in the lab frame.
- ⑤ (4 points) Write out the lagrangian and the corresponding equations of motion for the system formed in final state. Beware that all particles move with relativistic velocities and have nonzero electric charges. For simplicity You may disregard the spins of particles.

Problem 1 - Solution

- After proton “absorbs” photon, the system has a 4-momentum $P = p_\gamma + p_p$ and later decays into $J/\psi, p$
- The derivation of energy-direction relation verbatim repeats what we did in Lecture 4 for $M \rightarrow m_1 + m_2$ particle decay, but we should adjust

$$M \rightarrow \sqrt{(p_\gamma + p_p)^2} = \sqrt{m_p^2 + 2E_\gamma m_p}, \quad m_1 \rightarrow m_{J/\psi}, \quad m_2 \rightarrow m_p$$

In the C-frame of J/ψ and p , the 2 particles move in opposite directions, and we get respectively for energies

$$\Rightarrow E_{J/\psi}^{(C)} = \frac{2E_\gamma m_p + m_{J/\psi}^2}{2\sqrt{m_p^2 + 2E_\gamma m_p}}, \quad E_p^{(C)} = \frac{2m_p^2 + 2E_\gamma m_p - m_{J/\psi}^2}{2\sqrt{m_p^2 + 2E_\gamma m_p}}$$

The velocity V of “C-frame” w.r.t. lab-frame may be found recalling that for “compound” moving particle “ M ”

$$\mathbf{V} = \frac{\mathbf{P}}{E} = \frac{\mathbf{p}_\gamma + \mathbf{p}_p}{E_\gamma + E_p} = \frac{E_\gamma}{E_\gamma + m_p} \hat{\mathbf{p}}_\gamma$$

where $\hat{\mathbf{p}}_\gamma$ is a unit vector in direction of γ .

Problem 1 - Examen Final

Evaluation of the relative velocity v_{rel} :

- By definition, this is velocity of one of the particles in the rest frame of the other

$$p_{M_1} = (M_1 c, 0), \quad p_{M_2} = \left(\frac{M_2}{\sqrt{1 - v_{\text{rel}}^2}}, \frac{M_2 \mathbf{v}_{\text{rel}}}{\sqrt{1 - v_{\text{rel}}^2}} \right),$$
$$\Rightarrow (p_{M_1} \cdot p_{M_2}) = \frac{M_1 M_2}{\sqrt{1 - v_{\text{rel}}^2/c^2}} \quad (3)$$

Solving (3), can get invariant relative 4-velocity:

$$v_{\text{rel}} = c \sqrt{1 - \frac{M_1^2 M_2^2}{(p_{M_1} \cdot p_{M_2})^2}} = c \sqrt{1 - \frac{4m_p^2 m_{J/\psi}^2}{(2E_\gamma m_p - m_{J/\psi}^2)^2}}$$
$$p_{M_1} \cdot p_{M_2} = \frac{(p_{M_1} + p_{M_2})^2 - p_{M_1}^2 - p_{M_2}^2}{2} = \frac{m_p^2 + 2E_\gamma m_p - m_p^2 - m_{J/\psi}^2}{2}$$
$$= \frac{2E_\gamma m_p - m_{J/\psi}^2}{2}$$

Problem 1 - Solution

- In what follows we'll assume that γ propagates in direction $\hat{x}$, and θ_a ($a = J/\psi, p$) is angle between momentum of particle and $\hat{x}$
- In lab frame the direction angle θ of each particle is related to its energy $E^{(\text{lab})}$, as found in L4, by

$$E_a^{(C)} = \frac{E_a^{(\text{lab})} - V p_a^{(\text{lab})} \cos \theta}{\sqrt{1 - V^2/c^2}}$$

$$\Rightarrow \cos \theta_a = \frac{E_a^{(\text{lab})} - E_a^{(C)} \sqrt{1 - V^2/c^2}}{V p_a^{(\text{lab})}} = \frac{E_a^{(\text{lab})} - E_a^{(C)} \sqrt{1 - V^2/c^2}}{V \sqrt{(E_a^{(\text{lab})})^2 - m_a^2}}, \quad a = J/\psi, p$$

If we know the scattering angle for proton in the lab-frame, we may find its energy solving the corresponding equation,

$$\begin{aligned} E_p^{(\text{lab})} &= \frac{E_p^{(C)} \sqrt{1 - V^2} - V \cos \theta \sqrt{(E_p^{(C)})^2 (1 - V^2) - m_p^2 (1 - V^2 \cos^2(\theta))}}{1 - V^2 \cos^2(\theta)} \\ &= \frac{(E_\gamma + m_p) (2m_p (E_\gamma + m_p) - m_{J/\psi}^2)}{2(E_\gamma + m_p)^2 - 2E_\gamma^2 \cos^2(\theta)} - \\ &\quad - \frac{E_\gamma \cos(\theta) \sqrt{m_{J/\psi}^4 - 4m_{J/\psi}^2 m_p (E_\gamma + m_p) + 4E_\gamma^2 m_p^2 \cos^2(\theta)}}{2(E_\gamma + m_p)^2 - 2E_\gamma^2 \cos^2(\theta)} \end{aligned}$$

Problem 1 - Solution

If J/ψ decays later into two muons, we may iteratively apply the procedure which we found earlier: consider decay into rest frame of J/ψ and then make boost to lab frame
–In rest frame of J/ψ we repeat what we did in Lecture 4 for $M \rightarrow m_1 + m_2$ particle decay, but we should adjust

$$M \rightarrow M_{J/\psi}, \quad m_1 \rightarrow m_\mu \approx 0, \quad m_2 \rightarrow m_\mu \approx 0$$

$$\Rightarrow E_\mu^{(C)} = \frac{M_{J/\psi} + m_\mu^2 - m_\mu^2}{2 M_{J/\psi}} = \frac{M_{J/\psi}}{2},$$

each muon has the same energy, half of J/ψ mass

–In lab frame the direction angle θ of each particle is related to its energy $E^{(\text{lab})}$ as

$$E_a^{(C)} = \frac{M_{J/\psi}}{2} = \frac{E_a^{(\text{lab})} - V_{J/\psi} p_a^{(\text{lab})} \cos \theta_{J/\psi}}{\sqrt{1 - V_{J/\psi}^2/c^2}} \quad a = \mu^+, \mu^-$$

Clearly the maximum and minimum correspond to $\theta_{J/\psi} = \pi, 0$ (muon emission in direction of motion or opposite), so we may solve this algebraic equation

$$\left(E_\mu^{(\text{lab})}\right)_{\text{max, min}} = \frac{M_{J/\psi} \pm V_{J/\psi} \sqrt{M_{J/\psi}^2 - 4m_\mu^2}}{2\sqrt{1 - V_{J/\psi}^2/c^2}},$$

Problem 1 - minimal energy

Earlier we found that in the C -frame, where $\mathbf{p}_{J/\psi} + \mathbf{p}_{p'} = 0$, the energies of recoil proton and J/ψ are given by

$$\Rightarrow E_{J/\psi}^{(C)} = \frac{2E_\gamma m_p + m_{J/\psi}^2}{2\sqrt{m_p^2 + 2E_\gamma m_p}}, \quad E_p^{(C)} = \frac{2m_p^2 + 2E_\gamma m_p - m_{J/\psi}^2}{2\sqrt{m_p^2 + 2E_\gamma m_p}}$$

Clearly, we must have $E_{J/\psi} > m_{J/\psi}$ and $E_p > m_p$ in any frame, so we may get solving these algebraic identities that

$$E_\gamma \geq \frac{m_{J/\psi} (m_{J/\psi} + 2m_p)}{2m_p} = m_{J/\psi} + \frac{m_{J/\psi}^2}{2m_p} \approx 2.5 m_{J/\psi}$$

Alternative derivation:

Assume we observe everything from the rest frame of recoil proton p , and meson J/ψ has momentum $\mathbf{p}_{J/\psi}$

—Mandelstam variable s :

$$s = (p_p + p_{J/\psi})^2 = (m_p + E_{J/\psi})^2 - \mathbf{p}_{J/\psi}^2 = m_p^2 + m_{J/\psi}^2 + 2m_p E_{J/\psi}$$

$\Rightarrow$ minimal value of s corresponds to J/ψ at rest in protons's rest frame.

As we'll see below, the minimal value of s corresponds to the minimal photon energy E_γ .

Problem 1 - minimal energy

We have seen in Lecture 4 that for $1 + 2 \rightarrow 3 + 4$ collision the variable of Mandelstam s

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2$$

should satisfy identities

$$s \geq (m_1 + m_2)^2, \quad s \geq (m_3 + m_4)^2,$$

In our case $(m_3 + m_4)^2 > (m_1 + m_2)^2$ so it is sufficient to consider only the second identity

▷ In the proton's rest frame $p_p^\mu = (m_p c, 0)$, and the photon's momentum is $p_\gamma^\mu = (E_\gamma/c, \mathbf{p}_\gamma)$ where $E_\gamma = c |\mathbf{p}_\gamma|$

$$\begin{aligned} s &= (p_\gamma + p_p)^2 = p_\gamma^2 + p_p^2 + 2p_\gamma \cdot p_p = m_p^2 + 2m_p E_\gamma \geq (m_p + m_{J/\psi})^2 \\ \Rightarrow E_\gamma &\geq \frac{(m_p + m_{J/\psi})^2 - m_p^2}{2m_p} = \frac{m_{J/\psi} (m_{J/\psi} + 2m_p)}{2m_p} = m_{J/\psi} + \frac{m_{J/\psi}^2}{2m_p} > m_{J/\psi} \end{aligned}$$

where the equality (minimal energy) corresponds to production of J/ψ with zero relative momentum with respect to recoil proton p .

–Due to momentum conservation, J/ψ and recoil proton are produced with nonzero momentum in lab frame, that's why $E_\gamma \gg m_{J/\psi}$

Problem 1 - Mandelstam variables

The Mandelstam variables are defined as follows (below we use notations 1=proton, 2= γ , 3= p' (recoil proton), 4= J/ψ):

–The invariant energy:

$$s = (k_1 + k_2)^2 = m_p^2 + 2m_p E_\gamma$$

–The invariant momentum transfer (to the proton):

$$t = (k_1 - k_3)^2 = 2m_p^2 - 2k_1 \cdot k_3 = 2m_p^2 - 2m_p \cdot E_p^{(\text{lab})}$$

where $E_p^{(\text{lab})}$ is the energy of recoil proton found earlier (as a function of E_γ and θ)

–Finally, the variable $u = (k_1 - k_4)^2$ can be found from

$$s + t + u = \sum_{a=1}^1 m_a^2 = \underbrace{m_\gamma^2}_0 + m_p^2 + m_p^2 + m_{J/\psi}^2$$

$$u = m_{J/\psi}^2 - m_p^2 - 2m_p E_\gamma + 2m_p \cdot E_p^{(\text{lab})} = m_{J/\psi}^2 - m_p^2 - 2m_p E_{J/\psi}$$

Problem 1 - Lagrangian

In the final state we have 3 charged particles: a proton p and 2 muons μ^+, μ^- . The classical lagrangian for this case is given by (see Lecture 12)

$$S = S_0 + S_{\text{int}} + S_{\text{field}}$$

$$S_0 = - \sum_{i=p, \mu^+, \mu^-} m_i c^2 \int d\tau_i = - \sum_{i=p, \mu^+, \mu^-} m_i c^2 \int dt \sqrt{1 - \frac{v_i^2}{c^2}}$$

$$S_{\text{int}} = - \sum_{i=p, \mu^+, \mu^-} q_i \int d\tau_i (u_i \cdot A) = - \sum_{i=p, \mu^+, \mu^-} q_i \int dt (u_i \cdot A)$$

and the electromagnetic field term $S_{\text{field}} = -\frac{1}{2} \int d^4x F_{\mu\nu} F^{\mu\nu}$

–No terms with long-range Coulomb interaction, only local interaction with the field
Equation of motion:

$$\frac{d\mathbf{p}_{(i)}}{dt} = q_i (\mathbf{E}(\mathbf{x}_{(i)}, t) + \mathbf{v}_{(i)} \times \mathbf{B}(\mathbf{x}_{(i)}, t))$$

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}$$

Subscript (i) emphasizes that we should take electric field in the point $\mathbf{x}_{(i)}$ where the particle (i) is located

Problem 2 - part 1

- ① The lagrangian of the nonrelativistic particle of mass m moving in the external 3D field is given by

$$L = e^{\gamma t} \left(\frac{m \dot{r}^2}{2} - \frac{\kappa r^2}{2} \right), \quad \kappa, \gamma = \text{const}, \quad \kappa > 0, \quad \gamma > 0. \quad (1)$$

- **(3 points)** Write out the Euler-Lagrange equations for this system and compare them with equations for $\gamma = 0$ (spherical harmonic oscillator). How do You expect the new term $\sim \gamma$ will affect the trajectories?

Problem 2 - part 1

Solution: Working in Cartesian coordinates, can get immediately

$$0 = \frac{\partial L}{\partial r_a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}_a} \right) = -e^{\gamma t} (m\ddot{r}_a + \gamma m\dot{r}_a + \kappa r_a) = 0$$

This is equation for damped oscillator in 3D, parameter γ controls damping:

General solution :

$$r_a = A_a e^{-\gamma t/2} \cos \left(t \sqrt{\frac{\kappa}{m} + \left(\frac{\gamma}{2} \right)^2} + \varphi_a \right)$$

Constants A_a , φ_a should be fixed from initial conditions

–For $\gamma = 0$ have oscillations with constant amplitude, for $\gamma \neq 0$ the amplitude of oscillations gradually decreases

Problem 2 - part 2

- Identify continuous symmetries of this system and the integrals of motion associated with them due to Noether's theorem. Analyze if there are continuous symmetries of type

$$t \rightarrow t + \alpha, \quad r \rightarrow r\beta, \quad \alpha, \beta = \text{const}$$

which do not change the action of the system. Demonstrate that all the integrals of motion are conserved using equations of motion.

- ▶ We have seen in lectures that for time-independent central field, the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid for the system described by lagrangian (1). Analyze if the second law of Kepler remains valid for this system.

Problem 2 - part 2 (solution)

- ① The lagrangian has spherical symmetry, i.e. rotations around arbitrary vector $\mathbf{n}$,

$$\mathbf{r} \rightarrow \mathbf{r} + \phi \underbrace{\mathbf{n} \times \mathbf{r}}_{\equiv \boldsymbol{\gamma}}, \quad \phi \ll 1, \quad \phi = \text{const}$$

Repeating derivation of angular momentum conservation from Lecture 6 (page 23), we may get that there is a conserving quantity

$$Q = \mathbf{p} \cdot \boldsymbol{\gamma} = \sum_a \frac{\partial L}{\partial \dot{q}_a} \gamma^a = \mathcal{M} \cdot \mathbf{n} = e^{\gamma t} \mathbf{n} \cdot [\mathbf{r} \times m\mathbf{v}], \quad \mathbf{p} = m\mathbf{v}e^{\gamma t}$$

i.e. the “classical” angular momentum $\mathbf{M} = \mathbf{r} \times m\mathbf{v}$ is not conserved, but instead we have conservation of angular momentum defined as $\mathcal{M} = \mathbf{r} \times \mathbf{p} = e^{\gamma t} \mathbf{M} = \text{const.}$ The variable $\mathbf{M}$ decreases exponentially as expected,

$$\mathbf{M}(t) = \mathcal{M} e^{-\gamma t}$$

Problem 2 - part 2 (solution)

- 1 The force acting on the particle

$$m\ddot{\mathbf{r}}_a = -\gamma\dot{\mathbf{r}}_a - \kappa\mathbf{r}_a$$

The force always remains in the same plane as vectors $\mathbf{r}$, $\mathbf{v}$, so the trajectory will remain in this plane forever (as in $\gamma = 0$ case):

Second law of Kepler **is NOT valid now**

- 1 Original formulation for $\gamma = 0$:

A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time

Proof:

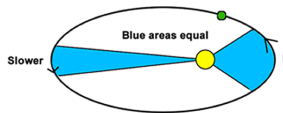
Based on conservation on angular momentum: $M_z \equiv$

$$p_\varphi = mr^2 \frac{d\varphi}{dt} = 2m \frac{dA}{dt} = \text{const}$$

- 2 For $\gamma \neq 0$ situation changes:

$$M_z(t) \equiv 2m \frac{dA}{dt} = \mathcal{M}_z e^{-\gamma t} \neq \text{const}, \quad A(t) = -\frac{\mathcal{M}_z}{2m\gamma} e^{-\gamma t}$$

i.e. the area which sweeps out a line segment decreases as a function of time



Problem 2 - part 2 (solution)

$$S = \int L dt, \quad L = e^{\gamma t} \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2} \right), \quad \kappa, \gamma = \text{const}, \quad \kappa > 0, \quad \gamma > 0$$

Can observe that the action is invariant under

$$t \rightarrow t + \alpha, \quad \mathbf{r} \rightarrow \beta \mathbf{r}, \quad \beta = e^{-\gamma \alpha/2}, \quad \alpha = \text{const}$$

To find Noether invariant, should assume $\alpha = \alpha(t) \ll 1$ and pick the terms $\sim \dot{\alpha}$ (the terms $\sim \alpha$ cancel)

–Caution with differentials:

$$dt \rightarrow dt(1 + \dot{\alpha}), \quad \dot{\mathbf{r}} \equiv \frac{d\mathbf{r}}{dt} \rightarrow \frac{\left(\dot{\mathbf{r}} - \frac{\gamma \dot{\alpha}}{2} \mathbf{r} \right)}{1 + \dot{\alpha}} + \mathcal{O}(\alpha)$$

$$\delta S = \dots = - \int dt \, \dot{\alpha} e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] + \mathcal{O}(\alpha)$$

Problem 2 - part 2 (solution)

Integral of motion :

$$Q = e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right]$$

For $\gamma = 0$ this is just (minus) energy; for $\gamma \neq 0$ energy is not conserved.
Explicit check that $Q = \text{const}$:

$$\frac{dQ}{dt} = -\frac{1}{2} e^{\gamma t} (2\dot{\mathbf{r}} + \gamma \mathbf{r}) \cdot (m(\ddot{\mathbf{r}}(t) + \gamma \dot{\mathbf{r}}(t)) + \kappa \mathbf{r}(t))$$

For trajectories which are solutions of the equations of motion

$$m\ddot{\mathbf{r}}_a = -m\gamma\dot{\mathbf{r}}_a - \kappa\mathbf{r}_a \Rightarrow dQ/dt = 0$$

General solution :

$$\mathbf{r}_a = A_a e^{-\gamma t/2} \cos \left(t \sqrt{\frac{\kappa}{m} - \left(\frac{\gamma}{2}\right)^2} + \varphi_a \right)$$

$$\Rightarrow Q = \frac{m}{2} \mathbf{A}^2 \left(\frac{\kappa}{m} - \left(\frac{\gamma}{2}\right)^2 \right) = \text{const}$$

Problem 2 - parts 3-4

- **(3 points)** Identify continuous symmetries of this system and the integrals of motion associated with them due to Noether's theorem. Analyze if there are continuous symmetries of type

$$t \rightarrow t + \alpha, \quad r \rightarrow r\beta, \quad \alpha, \beta = \text{const}$$

which do not change the action of the system. Demonstrate that all the integrals of motion are conserved using equations of motion.

- ▶ We have seen in lectures that for time-independent central field, the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid for the system described by lagrangian (1). Analyze if the second law of Kepler remains valid for this system.
- **(3 points)** Write out the generalized momenta and construct the hamiltonian
 - ▶ **(3 points)** Write out the canonical equations of motion. Demonstrate that they are equivalent to equations of Euler-Lagrange.

Problem 2 - parts 3-4 (solution)

- Generalized momenta and hamiltonian

$$p_a = \frac{\partial L}{\partial \dot{r}_a} = m\dot{r}_a e^{\gamma t}, \quad H = p_a \dot{r}_a - L = \left(\frac{\mathbf{p}^2}{2m} e^{-\gamma t} + \frac{\kappa r^2}{2} e^{\gamma t} \right)$$

- Canonical equations of motion:

$$\dot{r}_a = \frac{\partial H}{\partial p_a} = \frac{p_a}{m} e^{-\gamma t}, \quad \dot{p}_a = -\frac{\partial H}{\partial r_a} = -\kappa r_a e^{\gamma t}$$

If we substitute $p_a \rightarrow m\dot{r}_a e^{\gamma t}$, get immediately

$$m(\ddot{r}_a + \gamma \dot{r}_a) e^{\gamma t} = -\kappa r_a e^{\gamma t}$$

in agreement with Euler-Lagrange.

Problem 2 - parts 5-6

- Assume that we make a transformation of coordinates and momenta $(\mathbf{r}, \mathbf{p}) \rightarrow (\mathbf{R}, \mathbf{P})$ given by $\mathbf{r} = \mathbf{R}e^{-\gamma t/2}$ and $\mathbf{p} = \mathbf{P}e^{\gamma t/2}$.
 - ▶ **(3 points)** Demonstrate that this is canonical transformation and find corresponding generating function (it is recommended to construct $F(\mathbf{r}, \mathbf{P}, t)$ since the analysis is simpler in this case)
 - ▶ **(3 points)** Find the hamiltonian $\tilde{H}(\mathbf{R}, \mathbf{P})$ after the transformation and demonstrate that it does not depend explicitly on time t . Pay attention that your canonical transformation and its generating function depend explicitly on time.
- **(2 points)** Write out the equation of Hamilton-Jacobi for this system.

Problem 2 - parts 5-6 (solution)

- The canonical transformation has a form

$$\mathbf{p} = \mathbf{P}e^{\gamma t/2}, \quad \mathbf{r} = \mathbf{R}e^{-\gamma t/2}, \quad \Rightarrow \quad \mathbf{R} = \mathbf{r}e^{\gamma t/2}$$

- ▶ Can check that $[P_a, P_b] = 0$, $[R_a, R_b] = 0$, $[P_a, R_b] = \delta_{ab}$
- Construct generating $F(\mathbf{r}, \mathbf{P}, t)$ which generates this transformation:

$$\frac{\partial F}{\partial r_a} = p_a = P_a e^{\gamma t/2}, \quad \frac{\partial F}{\partial P_a} = R_a = r_a e^{\gamma t/2},$$

Straightforward integration yields $F(\mathbf{r}, \mathbf{P}, t) = \mathbf{r} \cdot \mathbf{P} e^{\gamma t/2}$

- ▶ Since $F(\mathbf{r}, \mathbf{P}, t)$ depends on time, this yields “extra” term in Hamiltonian transformation,

$$\tilde{H} - H = \frac{\partial F}{\partial t} = \frac{\gamma}{2} \mathbf{r} \cdot \mathbf{P} e^{\gamma t/2} = \frac{\gamma}{2} (\mathbf{R} \cdot \mathbf{P})$$

so the hamiltonian after transformation is

$$\tilde{H} = \frac{\mathbf{P}^2}{2} + \frac{\kappa \mathbf{R}^2}{2} + \frac{\gamma}{2} (\mathbf{R} \cdot \mathbf{P})$$

Problem 2 - parts 5-6 (solution)

- The Hamilton-Jacobi equation is (in terms of variables $\mathbf{p}, \mathbf{r}$)

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} e^{-\gamma t} + \frac{\kappa r^2}{2} e^{\gamma t} = 0.$$

- ▶ Has explicit dependence on time, so integration is not straightforward
- If we use “new” variables $\mathbf{P}, \mathbf{R}$, get the Hamilton-Jacobi equation is

$$\frac{\partial S}{\partial t} + \frac{(\nabla_{\mathbf{R}} S)^2}{2m} + \frac{\kappa R^2}{2} + \frac{\gamma}{2} (\mathbf{R} \cdot \nabla_{\mathbf{R}} S) = 0.$$

No explicit t -dependence implies can use separation of variables and get $S = -Et + f(R_1, \dots, R_N, \alpha_1, \dots, \alpha_N)$ as usual

Problem 2 - [part 2++]

Integrals of motion (symmetry generators):

- The invariant Q which we found earlier may be rewritten as

$$Q = e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] = \left[\left(\frac{\mathbf{p}^2}{2m} e^{-\gamma t} + \frac{\gamma}{2} \mathbf{p} \cdot \mathbf{r} \right) + \frac{\kappa r^2}{2} e^{\gamma t} \right]$$

- Components of vector $\mathcal{M}_a = [\mathbf{r} \times m e^{\gamma t} \dot{\mathbf{r}}] = \mathbf{r} \times \mathbf{p}$
- Poisson theorem: if f and g are integrals of motion, then $h = [f, g]$ is also integral of motion
 - ▶ Can use to recover “new” integrals of motion
 - ▶ Straightforward evaluation of Poisson brackets:

$$[M_a, M_b] = -\epsilon_{abc} M_c, \quad [Q, M_a] = \dots = 0$$

there is no new invariants, algebra is closed.

Problem 2 - [final comment]

We could also replace variable

$$\mathbf{r}(t) \rightarrow e^{-\gamma t/2} \mathbf{R}(t)$$

in order to eliminate $e^{\gamma t}$ in lagrangian, so we would get:

$$L = \frac{m}{2} \dot{\mathbf{R}}^2 - \frac{m\gamma}{2} \mathbf{R}(t) \cdot \dot{\mathbf{R}}(t) - \frac{1}{2} \left(\kappa - \frac{m\gamma^2}{4} \right) \mathbf{R}^2$$

- No explicit dependence on time (only coordinates and velocity), so “energy” $\mathcal{E}$ is conserved, and coincides with Q and $\tilde{H}$

$$\mathcal{E} = \tilde{H} = \sum_a \dot{R}_a \frac{\partial L}{\partial \dot{R}_a} - L = \frac{m}{2} \dot{\mathbf{R}}^2 + \frac{1}{2} \left(\kappa - \frac{m\gamma^2}{4} \right) \mathbf{R}^2 = -Q$$

$$Q = -e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] = -\frac{m}{2} \dot{\mathbf{R}}^2 - \frac{1}{2} \left(\kappa - \frac{m\gamma^2}{4} \right) \mathbf{R}^2$$

where $\mathbf{P} = \partial L / \partial \dot{\mathbf{R}} = m\dot{\mathbf{R}} - m\gamma \mathbf{R}/2$ and $\tilde{H} = \frac{\mathbf{P}^2}{2m} + \frac{\kappa \mathbf{R}^2}{2} + \frac{\gamma}{2} (\mathbf{R} \cdot \mathbf{P})$